Zuzanna Stachlewska

/zuˈzannːa/ /staˈxlɛfska/

Personal profile

Materials engineer with over 3 years of experience as materials researcher and a Master's degree in Materials and Manufacturing Engineering from DTU, driven by sustainable, well-designed and high-quality engineering solutions. I specialize in surface engineering and materials investigation, working with SEM/EDS, XRD, GD-OES, LOM, metallography and mechanical testing. Strong science communicator who enjoys tailoring complexity to the audience. Analytical, detail-oriented and proudly a transgender woman who values inclusivity and good collaboration. Outside the lab, you'll find me cycling through North Zealand's forests and hills or capturing details of the surrounding world with a camera in hand.

Experience

Materials and Surface Specialist - Expanite A/S

2023-01 – Present

Achievements:

- Developed and delivered to the market a novel surface hardening solution with potential multi-milion DKK annual revenue. [1,2,3,4,6,8,10,13,14]
- Developed a surface hardening solution for an automotive customer, currently in ramp-up phase toward high-volume production. [1,2,3,4,6,8,10,13,14]
- Reduced process times by 5% through optimisation of furnace cycles. [4,6,10]

Key responsibilities:

- **Leading customer projects** [13] in surface hardening or heat treatment, including vendor coordination.
- **Performing R&D** [1,2,3,4,6,8,10,13,14] activities, including qualitative and quantitative laboratory analyses.
- **Conducting metallurgical investigations** [1,2,3,6], including sample preparation, light optical microscopy (LOM), and hardness analysis.
- **Analyzing results** [6,11,14] and communicating findings to internal and external stakeholders.
- **Planning, executing, and evaluating tests** [1,6,8,11,13] for core products and processes.
- **Troubleshooting** [6,10,14] vacuum systems (furnaces) and analyzing heat treatment process data.
- **Investigating non-conformities** [14] and customer claims using root cause analysis (RCA) methodologies.
- **Providing technical support** [3,6,11] to Production, Sales, and Quality teams.
- **Designing and implementing** [6,7,8,13] technical fixtures and vacuum analysis auxiliary systems.
- **Managing and creating technical documentation:** [10,11] PPAP, SOP, and QMS documentation.
- **Writing** [11] whitepapers and creating technical materials for marketing team.

Information

Pronouns:
She/Her

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Telephone number:
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Date of birth:
1997-05-15

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Klostervej 31, 3.5
3400 Hillerød
Relocation to Aarhus is possible

Hard skills

[1] Materials characterisation
[2] Surface engineering
[3] Metallurgy
[4] Heat treatment
[5] Additive manufacturing
[6] Data analysis
[7] Technical design (CAD)
[8] Experiment design
[9] PVD hard coatings

Soft skills

[10] Problem-solving
[11] Communication
Communicating complex ideas to various stakeholders with tailored level of complexity
[12] Life-long learner
[13] Project management
Managing multiple projects across organisation
[14] Analytical skills
Applying RCA methodologies to solve QA tasks and equipment malfuncions

Education

2020–09 to 2022–08

Technical University of Denmark (DTU) – Faculty of Mechanical Engineering

MSc in Materials and Manufacturing Engineering

Thesis: Additive manufacturing of lead-free (KNa)NbO_3 piezoelectric ceramics with the feasibility study of a templated growth approach. [1,5,6,7,8]

Best project award: Design of hearing aid battery compartment utilizing AM fast prototyping approach (Micro product design, development and production - DTU: 41743). [5,7,10]

2016–09 to 2020–07

Lodz University of Technology (TUL) – Faculty of Mechanical Engineering

BSc Eng. in Materials Engineering

Thesis: Metal-Ceramic Composite in a SiC/Inconel 617 system as a potentially new material for brake systems. [1,2,3,4,6,8,9]

Scholarship: Awarded with a scholarship granted to 10% of best ranking students.

2019–02 to 2019–07

Università degli studi di Napoli Federico II (UNINA) – Dipartimento di Ingegneria Chimica, dei Materiali e della Produzione Industriale (Erasmus+)

MSc level courses during BSc Eng. programme

Courses conducted entirely in Italian.

Software skills

Data Analysis [6]
OriginLab / Fiji (ImageJ) / Excel

Technical writing tools [11]
LaTeX (Overleaf) / Typst

Microsoft Office package
Excel / Word / PowerPoint

Computer Aided Design (CAD) [7]
FreeCAD / SOLIDWORKS

Programming [12]
Python (GUI app, GitHub repository [Link]) / Matlab

Visual communication [11]
Adobe Illustrator / Adobe Photoshop / Inkscape / Affinity

English
Full professional proficiency (8.0 IELTS score, 2020)

Danish
Intermediate (PD2 certificate, 2023)

Polish
Native

Italian
Intermediate